

Hemal Rajput

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Education

McMaster University, BEng in Software Engineering

Expected May 2027

- **GPA:** 3.73/4.0
- **Minor:** Biochemistry
- **Relevant Electives:** Natural Language Processing, Predictive & Intelligent Control, Bioinformatics

Work Experience

Data Scientist and Artificial Intelligence, Canadian Space Agency – Longueuil, QC

Sept 2025 - Present

- Refactored three department-specific **retrieval-augmented generation assistants** into a configuration-driven **Flask and JavaScript platform**, centralizing, document-access, and UI logic so shared improvements propagate across applications
- Designed and implemented a modular **ingestion pipeline**; supported application-isolated runs, incremental change detection, targeted single-document processing, forced reprocessing, and source reconciliation
- Architected isolated **Azure Blob Storage and Azure AI Search** knowledge bases for three internal departments, implementing a common vector-search schema and dual ingestion paths for pipeline-generated JSON and Azure-managed raw-document processing
- Developed an **LLM evaluation framework** comparing assistant responses against human reference answers using lexical similarity, semantic similarity, entailment, faithfulness, and LLM-as-judge metrics; presented the methodology and findings at the CSA's Community of Practice
- Contributed to the CSA's **AI governance framework**, translating hands-on LLM development experience into guidance for responsible AI evaluation, data governance, risk management, and secure deployment

Software Developer, Civiconnect – St. Catharines, ON

2024

- Built a bespoke restaurant procurement website that brought ordering from multiple suppliers into one workflow using React, Next.js, Tailwind CSS, and PostgreSQL
- Designed the PostgreSQL schema in Supabase with foreign key constraints and implemented authentication and row-level security for the ordering system

Projects

Waypoint – Mobile Robot Tracking and State Estimation, C++17, Python

[GitHub](#)

- Implemented a C++ pure-pursuit controller and planar EKF fusing wheel-encoder and gyroscope measurements; evaluated differential-drive tracking with ground-truth, odometry, and EKF feedback
- Built reproducible Python experiments across routes and disturbance scenarios, measuring tracking and localization error, endpoint accuracy, and success under encoder noise, wheel mismatch, wheel slip, and gyro bias

PronouncedFish – AI-Assisted Speech Practice

Capstone – Design Phase

- Collaborating with a five-person team to design a speech-practice platform with pronunciation API feedback and clinician progress tracking; proposed a React, FastAPI, and PostgreSQL architecture

Open-Source Space Data Tutorials, Python – CSA

[THEMIS](#) | [CARISMA](#)

- Published two bilingual Jupyter Notebook tutorials on the CSA's official GitHub, covering auroral image processing and animation, plus parsing and visualization of magnetometer and riometer data

Leadership & Service

NASA International Space Apps Challenge, CSA Mentor & Judge

2025–2026

- Guided teams in using Canadian space datasets and evaluated project submissions against challenge-specific judging criteria

Technologies

Languages: Python, Java, C++, JavaScript, SQL

Frameworks & Libraries: Flask, React, Next.js, Hugging Face Transformers, OpenCV, Tailwind CSS

Cloud & Databases: Azure AI Search, Azure Blob Storage, Azure App Service, Azure AI Foundry, PostgreSQL, Supabase

Tools: Git, Docker, Linux, MLflow, Jupyter